MIL-T-37318
22 June 1976

MILITARY SPECIFICATION

TEST TUBE, 16 by 125 mm., Screw Cap, Disposable

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a 16 x 125 millimeter (mm.), disposable glass culture tube with screw cap closure, suitable for autoclaving.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

DD-G-541	Glass, Laboratory.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.

Military

MIL-P-116	Preservation-packaging, Methods of.
MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 6640

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2. Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Designation C148 Polariscopic Examination of Glass Containers.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Glass. The test tube shall be fabricated from borosilicate glass that conforms to DD-G-541, Type I, Class A or B.

3.1.2 Plastic. The screw cap shall be fabricated from polypropylene that meets the requirements hereinafter indicated.

3.2 Design.

3.2.1 Test tube. The test tube shall be cylindrical, and shall have an open end and a closed, rounded bottom. The open end of the test tube shall have a continuous type exterior thread. The test tube shall conform to Table I for dimensions:

Table I Dimensions of test tube

Characteristic	Dimensions (mm.)
Overall length	125 ± 2.5
Outside diameter at center of tube	15.25 minimum
Inside diameter at open end	9.25 minimum

When tested as specified in 4.4.1, the capacity of the test tube shall be not less than 15 milliliters (ml.). When tested as specified in 4.4.2, the ring strain shall

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be not greater than that of temper no. 1; any longitudinal strain shall be faint and highly diffused.

3.2.2 Screw cap. The screw cap shall be a one-piece linerless design. The inner surface of the screw cap shall be sufficiently threaded with a continuous type thread for proper closure on the test tube. In closure, the screw cap shall form an effective seal with the top surface of the test tube to prevent the loss of liquid contents (see 3.3.1).

3.3 Performance.

3.3.1 Leakproof closure. When tested as specified in 4.4.3, there shall be no evidence of leakage from the test tube.

3.3.2 Autoclavability. When tested as specified in 4.4.4, the test tube and screw cap closure shall not deform or crack.

3.4 Workmanship. Workmanship shall be first class throughout. The test tube and screw cap shall be free of defects which detract from their appearance or may impair their serviceability. The test tube shall be symmetrical in shape, and free from striae, checks, chips, cracks and bubbles. The top surface at the opening of the test tube shall be level and free of ridges. The test tube shall be dry, clean, clear and ready for use; it shall be free from grease, dirt, stains and other foreign matter. The screw cap shall be free of sharpness, deterioration, burrs, cracks, chips and breaks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table II. The unit of product for examination purposes shall be the individual test tube or screw cap as applicable. No more than 10 test tubes or screw caps shall be selected from a unit package (see 5.1.1).

Table II Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table III. The unit of product for testing purposes shall be the individual test tube. The acceptance number shall be zero. No more than 3 test tubes shall be selected from a unit package (see 5.1.1).

Table III Sampling for tests

Component	Characteristic	Requirement	Test procedure	Inspection level
Test tube	Capacity	3.2.1	4.4.1	S-1
Test tube	Strain	3.2.1	4.4.2	S-1
Test tube (w/cap)	Leakproof closure	3.3.1	4.4.3	S-1
Test tube (w/cap)	Autoclavability	3.3.2	4.4.4	S-1

1.3 Examination. The test tubes and screw caps shall be examined to determine compliance with the requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with Table IV, but shall not be restricted to the classified possible defects listed in the table.

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Table IV Classification of defects

Categories	Defects
(Bottle or screw cap)	
Major	
101	Test tube not clean, dry, clear and ready for use.
102	Test tube not free of checks, chips and cracks.
103	Test tube not free of rough and sharp edges.
104	Test tube not free of blisters.
105	Test tube not free of grease, dirt, stains and other foreign matter.
106	Screw cap not free of rough and sharp edges.
107	Screw cap not free of burrs, cracks, chips and breaks.
108	Thread on test tube or screw cap not complete, continuous and suitable for intended purpose.
109	Top surface at opening of test tube not level and free of ridges.
Minor	
201	Test tube not free of striae.
202	Test tube not free of bubbles.

4.3.2 Dimensional examination. The test tube shall be examined for defects in dimensions. Any dimension not within specified limits shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Capacity. The test tube shall be completely filled with water (i.e., to overflow capacity). The volume of water used to fill the test tube shall be measured and recorded.

4.4.2 Strain. The test tube shall be tested for strain in accordance with ASTM C148.

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4.4.3 Leakproof closure. The test tube shall be filled with water, and the screw cap shall be applied handtight. The test tube shall be placed in an inverted position and allowed to stand for 30 minutes before examining for evidence of leakage beyond the seal.

4.4.4 Autoclavability. The test tube with screw cap applied shall be autoclaved at 15 pounds per square inch (250° Fahrenheit) for 30 minutes. After autoclaving, the test tube shall be allowed to cool at room temperature and then examined.

5. PACKAGING.

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package, containing 250 test tubes as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Unit package. Each unit of issue shall be packaged in accordance with MIL-P-116, method III as follows: 250 tubes shall be packaged vertically in an open top fiberboard box having a minimum bursting strength of 125 pounds. Two hundred fifty caps shall be packaged in a plastic bag and placed at one end of the box containing the tubes. The box shall form a snug fit around the tubes and caps. The tubes shall extend approximately 1/2 inch above the top of the box, and closure of the box shall be effected. As an alternate, 250 tubes with caps loosely applied shall be packaged in a snug-fitting fiberboard box having a minimum bursting strength of 125 pounds. The box shall contain full-height chipboard partitions of suitable weight. The partitions shall form a snug-fitting individual cell for each tube. Closure of the box shall be adequate to prevent accidental opening by a 3 mil plastic shrink wrap.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A. Four units of issue shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. Four units of issue shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129. In addition, the exterior container shall be marked with the legend:

"GLASS--HANDLE WITH CARE."

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following item as listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-604-1102	TEST TUBE, 16 by 125 mm. Screw Cap, Disposable, 250s

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6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
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